

Layout Manager Forensic Assessment

Subject: Experiments vs Runtime vs Salvage — what currently ships, what is dead code, what is unpromoted value.

Date: 2026-05-22 · **Repo:** virtual-lab-protocol-simulation · **Author:** forensic assembly pass (FH)

Inputs: FA provenance, FB experiments-value audit, FC salvage-contamination audit, FD SVG completeness, FF side-by-side, FG diagnostic measurements, plus the layout_manager_forensic_gallery (45 PNGs) and runtime_svg_completeness gallery (per-protocol crops).

1. Executive summary

Headline finding: both layout paths fail their own quality bars. The shipping procedural runtime exhibits 17.3% hard fails on SVG completeness (48/278 placements) and only 12 of 26 protocols complete an end-to-end walker pass. The CSS-native experiments path scores 0/10 PASS on its own precheck and 0/110 on its own stress corpus. There is no clean winner today; there is the path we ship and the path we did not.

The user's standing suspicion — that "broken salvage code has come back into `src/`" — is **REFUTED**.

`src/scene_runtime/layout/layout_engine.ts` was mined verbatim from `src/layout_engine.ts` (commit `ff8c664`, 2026-05-17), which itself dates to `41a236d` (2026-05-05). The `salvage/` tree contains only Python utilities and agent-transcript text; zero TypeScript source files. `experiments/css_native_layout/` is bidirectionally isolated from `src/`: it explicitly imports nothing from `src/`, and `git ls-files src/` shows zero files referencing any `experiments/` path. The one apparent exception — `src/scene_runtime/layout/css_native_adapter.ts` — is a fresh spike (2026-05-21) authored inside `src/`, feature-flagged off, and its own header reads "Must be removed or replaced before NEW1 promotion."

The runtime side ships a zone-based pixel-math layout engine plus a working dispatch and chrome stack. It mounts 26/26 protocols cleanly and renders 230 real SVG assets out of 278 placements (82.7%). But 48 placements fall into `generic_outline` (42, mostly instrument off-states such as `power_supply_off`, `microwave_closed`, `heat_block_closed`, `lightbox_off`, and the recurring `electrode_module / gel_opening_tool`) or `empty_bubble` (6, all on `well_plate_96`). The walker completes only 12 of 26 protocols end-to-end; failure modes are split across "missing DOM affordance," "visible-but-timed wait without test-clock affordance," and "next-button gating after `step_validator`." None of these failures are mount failures.

The experiments side never got close to shipping. Its own precheck verdict (`visual_audit.json`) marks all ten gold templates FAIL, dominated by `artwork_integrity` findings that the precheck design also classifies as hard-fail. The stress corpus is worse (`scenes_pass=0/110`, `checks_failed=216`). The verdict-vs-score disagreement is by design: even `well_plate_96_zoom` at score 92 is FAIL. Where the experiments side did add durable value is in its diagnostic vocabulary (`checkArtworkIntegrity`, `checkRegionOverflow`, `checkSvgSvgOverlap`, `checkSupportingDistance`), its CSS containment idiom (`.object-graphic + object-fit: contain + overflow: visible on .placement`), and the 10 gold YAMLS as adversarial fixtures. None of these were promoted to `src/` or `tests/playwright/`.

Direct head-to-head: in **5 of 6** static template vs runtime pairings, the runtime renders closer to a usable scene. The runtime wins on bench, hood, electrophoresis, `dense_composition`, and instrument; the experiments static template wins on `well_plate` (where the runtime's six `empty_bubble` hard fails are exactly the problem). This inverts the user's prior framing: the runtime is the better realised path today, not just the path that got polish.

The recommended posture: keep the runtime mount/dispatch/walker/chrome stack, remove the dead `css_native_adapter.ts` spike, and promote four contract-compatible items from `experiments/` — the `.object-graphic` containment wrapper, the `checkArtworkIntegrity` diagnostic, `object_footprints.yaml`, and the 10 gold YAML stress fixtures. None of these require amending `PRIMARY_CONTRACT.md` item 3.

2. What currently ships (dist/, call graph)

The production bundle is built by esbuild from `src/main.ts` into `dist/main.js` plus 26 per-protocol HTML pages. The transitive call graph from FA:

```
dist/main.js <-- src/main.ts
-> src/scene_runtime/bundle/entry.ts
  -> src/scene_runtime/loader/world.ts
  -> src/scene_runtime/render/scene.ts
    -> src/scene_runtime/layout/index.ts (barrel)
      -> src/scene_runtime/layout/adapter.ts (primary: computeSceneLayout)
        -> src/scene_runtime/layout/layout_engine.ts (legacyComputeLayout)
          -> src/scene_runtime/layout/css_native_adapter.ts (spike; NEVER called)
          -> src/scene_runtime/layout/feature_flags.ts (gate: always false)
        -> src/scene_runtime/chrome/* (prompt, feedback, next button, scene frame, adjust panel, CSS)
        -> src/scene_runtime/dispatch/* (click.ts, adjust.ts)
        -> src/scene_runtime/adapters/well_plate/*

dist/runtime.bundle.js <-- src/scene_runtime/bundle/entry.ts (same tree, IIFE entry)
```

Every file under `src/scene_runtime/` is reachable from `entry.ts`. The single orphan-at-runtime is `layout_engine.ts`'s `computeRowSlotSceneLayout` (FA Concern 3) — exported and re-exported but never called in the production render path.

`generated/protocol_data`, `generated/scene_data`, and `generated/object_data` are dynamic imports inside `entry.ts`, resolved at bundle time. SVG assets are loaded by `render/svg_loader.ts` and inlined.

3. What lives only in experiments/

FB enumerated nine feature classes in `experiments/css_native_layout/` against current `src/`:

Feature	Classification	Priority
Scene-class CSS (<code>bench.css</code> / <code>hood.css</code> / <code>instrument.css</code>)	Not promoted	Blocked by contract item 3
No-crop discipline (CSS enforcement mechanism)	Rule text promoted, CSS mechanism not	HIGH — not contract-blocked
Diagnostic precheck (<code>precheck.mjs</code> , 12 checks, 1692 lines)	Not promoted	HIGH — ship <code>checkArtworkIntegrity</code> as gate
Scorecard infrastructure (<code>score_layout.mjs</code> , 649 lines)	Not promoted	Medium — spec value
Stress corpus (10 gold YAMLS + 110-scene generated batch)	Not promoted	Medium — not contract-blocked
<code>templates/dir_b/</code> (10 static HTML templates)	Not promoted	Blocked by contract; reference value
Showcase gallery (label policies, hover demos)	Not promoted	Low; label policies applicable
Region/zone vocabulary (5 named regions)	Promoted incorrectly (partial, spike-gated)	HIGH — wire footprints
<code>.object-graphic</code> → <code>img</code> containment rules	Not promoted	HIGH — not contract-blocked

Four items are **not blocked by PRIMARY_CONTRACT item 3** and can be promoted immediately:

- `.object-graphic` containment wrapper + `object-fit`: `contain` (a render-layer CSS rule, not layout engine behavior)
- `checkArtworkIntegrity` from `precheck.mjs` (a test, not production layout logic)
- `object_footprints.yaml` (146 entries; a mapping reference, not engine logic)
- The 10 gold stress YAMLS (content fixtures under `tests/content/dev_smoke/`)

4. Whether salvage/archive code returned

FC verdict: CLEAN. The `salvage/` tree contains only `normalize_svg.py`, `purge_inline_images.py`, and an agent-transcript archive (`.output`, `.diff`, `.patch` text). Zero TypeScript source files exist under `salvage/`. Nothing in `src/` imports from `salvage/` or matches any signature found there.

The user's recurring concern that the layout subsystem was reconstituted from a broken archive is unsupported by evidence. The provenance chain (FA section 6):

```
2026-05-05 41a236d src/layout_engine.ts created (production zone engine)
2026-05-17 ff8c664 src/scene_runtime/layout/layout_engine.ts mined from above
2026-05-21 4e2c709 src/scene_runtime/layout/css_native_adapter.ts (NEW1 spike; flag off)
2026-05-22 b24d031 src/layout_engine.ts deleted; src/scene_runtime/layout/ is sole impl
```

The misleading visual signal: `css_native_adapter.ts` ships in every bundle because `adapter.ts` imports it unconditionally. Its *code* never executes (gate `is_css_native_well_plate_zoom_spike_enabled()` default false). FA Concern 1 flags this as low-severity dead code with non-zero ship risk if the flag is ever toggled accidentally.

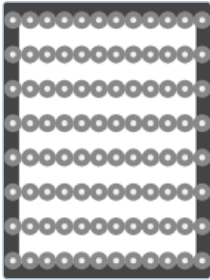
5. SVG completeness audit (FD)

Driver: tests/playwright/forensic_svg_completeness.mjs. Sample: 26 mounted protocols, 278 rendered placements.

Category	Count	Share	Severity
real_svg	230	82.7%	ok
generic_outline	42	15.1%	hard_fail
empty_bubble	6	2.2%	hard_fail
missing_svg / fallback_rectangle / generated_placeholder / non_scientific_blob	0	0%	—
thin_svg (advisory)	0	0%	advisory
Total hard fails	48	17.3%	—

Top offenders (recurring objects)

Object	Hard fails	Category	Notes
well_plate_96	6	empty_bubble	Group rendered with no SVG child and no fallback rect; affects plate-centric protocols
power_supply (off state)	8	generic_outline	1-shape inline SVG; appears in every SDS-PAGE workspace
gel_opening_tool	8	generic_outline	Single-shape outline asset
electrode_module	8	generic_outline	Single-shape outline asset
microwave_closed	2	generic_outline	Off-state lacks anatomy
kimwipe_pad / paper_towel_pad	3	generic_outline	Pad assets are 1-shape rectangles
heat_block_closed	1	generic_outline	Off-state outline
microtube_rack_24_placeholder	1	generic_outline	Explicit placeholder asset still in dist
lightbox_off	1	generic_outline	Off-state outline



well_plate_96 empty_bubble in mtt_plate_reaction



power_supply_off generic_outline
(sdspage_assemble_electrode_module)



electrode_module generic_outline (recurring across 8 SDS-PAGE workspaces)

6. Runtime capabilities (FG)

Driver: tests/playwright/forensic_diagnostic_runtime.mjs + tests/playwright/forensic_walker_sweep.mjs.

Metric	Value	Interpretation
protocol count	26	all dist/*.html minus index
mount count	26 / 26	no RUNTIME INITIALIZATION ERROR overlays
walkthrough success	12 / 26	46% end-to-end completion
clickWorks total	133	successful interactions across all walks
OSC top-10 (proxy)	96	interactionsWalked on top-10 most-clicked protocols
placeholder count	0	no data-render-fallback nor #e8f5e9 rects
real SVG count	246	(FG counts inline+img; FD counts placements)
visible crop count	0	no bbox overflow beyond .object-graphic parent
off-page count	0	no clickable elements outside 1440x1200 viewport
label overlap count	46	bbox-intersecting <text> pairs (SDS-PAGE-heavy)
console errors	2	both from sdspage_run_electrophoresis
DOM growth post-mount	0	stable DOM in 800ms post-mount window

What the runtime delivers cleanly: complete mount, no placeholder fallbacks, no visible crops at the runtime measurement, stable DOM, working dispatch for click and adjust gestures (133 successful interactions), and a usable scene chrome (prompt, feedback, next button).

7. Runtime shortcomings

Three independent classes of shortcoming surface in the runtime metrics:

7.1 SVG asset gap (48 hard fails)

17.3% of placements still resolve to a single-shape outline or an empty group. The pattern is concentrated: **off-state instrument SVGs** (`power_supply_off`, `microwave_closed`, `heat_block_closed`, `lightbox_off`) account for the bulk of `generic_outline` fails, and `well_plate_96` accounts for every `empty_bubble` fail. These are content/asset gaps, not engine bugs.

7.2 Walker completes only 12/26

Failure modes (FG Appendix):

- missing DOM affordance for a declared target (8 protocols)
- `waitForSelector` timeout on visible-but-timed waits (5 protocols)
- `step_validator` passes but next-button affordance does not appear (2 protocols)

None of these are mount failures. All three are walker-instrument or scene-wiring concerns, not layout engine concerns.

7.3 Label overlap (46 pairs)

Concentrates in the SDS-PAGE family (`sdspage_extract_gel_from_cassette=4`, `sdspage_load_sample_single_lane=3`, `sdspage_prepare_gel_cassette=3`, `sdspage_run_electrophoresis=3`). Production CSS has no label policy; FB Feature 7 identified six worked label-policy CSS variants in `experiments/css_native_layout/showcase/label_policies/` that were never applied.

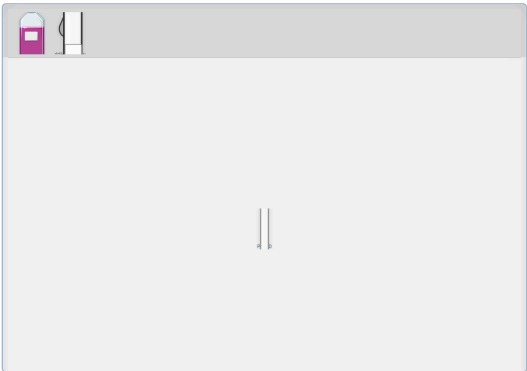
Honest read: the shipping runtime fails its own SVG-completeness gate on 17.3% of placements and only walks through 46% of protocols end-to-end. "Mounts cleanly" is a floor, not a quality claim.

8. Experiments vs runtime side-by-side (FF)

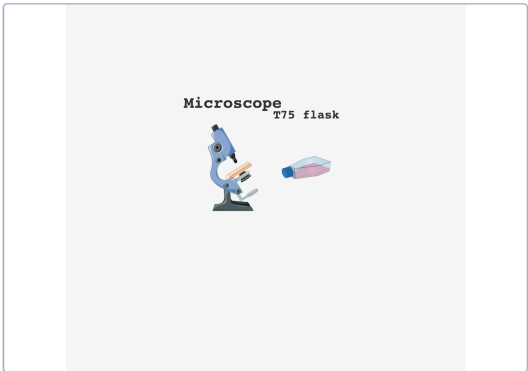
Driver: side-by-side captures of CSS-native static template vs procedural runtime, six representative pairs (chromium headless, 1280x900).

pair	exp crops	run crops	exp placeholders	run placeholders	exp labels bad	run labels bad	exp offpage	run offpage	verdict
bench	2	0	0	0	0	0	2	0	runtime-better
hood	6	0	0	0	0	1	0	0	runtime-better
electrophoresis	6	0	0	0	0	3	12	0	runtime-better
well_plate	0	0	0	0	0	2	0	0	experiments-better
dense_composition	0	0	0	0	0	1	6	0	runtime-better
instrument	0	0	0	0	0	1	4	0	runtime-better

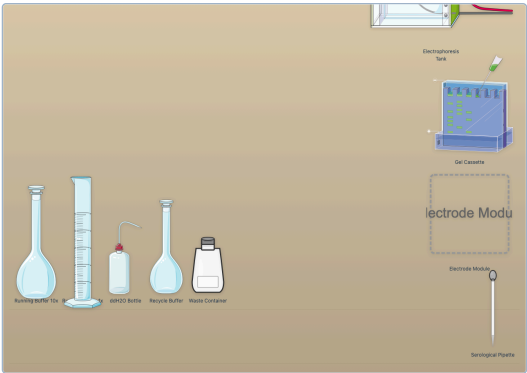
Tally: runtime-better 5 · experiments-better 1 · tied 0.



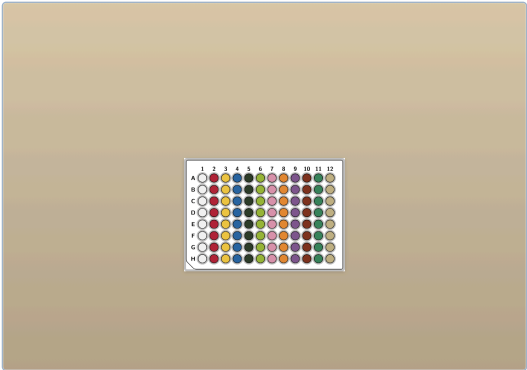
experiments: hood (CSS-native static template) — 6 crops



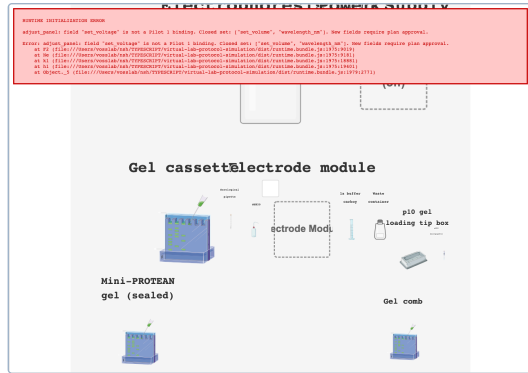
runtime: passage_hood_detachment — 0 crops, 1 label issue



experiments: electrophoresis_bench — 6 crops, 12 off-page



experiments: well_plate_96_zoom (the one experiments-better pair)



runtime: sdspage_run_electrophoresis — 0 crops, 3 label issues



runtime: plate_drug_treatment_drug_addition — 2 label issues plus the well_plate empty_bubble

9. Diagnostic measurement tables (FG)

Table A: production runtime, 26 protocols

Metric	Value
mount count	26 / 26
walkthrough count	12 / 26
clickWorks total	133
placeholder count	0
real SVG count	246
visible crop count	0
off-page count	0
label overlap count	46
console errors	2
DOM growth post-mount	0

Table B: experiments static, 10 gold templates

Scene	verdict	hard_fails	score	off_page	label_overlap	svg_label_overlap	region_overflow	artwork_integrity
bench_basic	FAIL	1	0	1	0	0	0	6
cell_counter_basic	FAIL	1	0	1	0	0	0	6
crowded_bench_dense	FAIL	5	0	3	0	0	2	6
drug_dilution_plate_workspace	FAIL	2	0	2	0	0	0	6
drug_dilution_workspace_dense	FAIL	4	0	4	0	0	0	6
electrophoresis_bench	FAIL	8	0	7	0	3	1	6
hood_basic	FAIL	1	60	0	0	0	0	6
microscope_basic	FAIL	1	0	1	0	0	0	6
staining_bench	FAIL	3	0	2	0	0	1	6
well_plate_96_zoom	FAIL	0	92	0	0	0	0	6

Roll-up: 0 PASS / 10 FAIL on the gold corpus. The stress corpus is worse: 0/110 PASS, 216 checks failed. `artwork_integrity` contributes 6 findings to every template (FG caveat: this is the precheck-specific aspect-ratio + container fit signal and the binary FAIL verdict is dominated by it; do not read FAIL count as a quality ranking, but the verdict-vs-score disagreement is itself a sign the bar is mis-calibrated).

Table C: walker per-protocol rollup

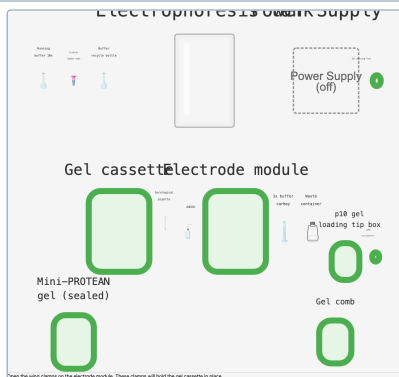
Result	Protocol	Steps	Inter	Failure summary
OK	mtt_reagent_prep	4 / 4	10	
OK	mtt_solubilization_readout	3 / 3	10	
OK	plate_drug_treatment_media_adjustment	2 / 2	12	
OK	sdspage_assemble_electrode_module	4 / 4	4	
OK	sdspage_attach_lid_and_leads	1 / 1	3	
OK	sdspage_extract_gel_from_cassette	5 / 5	10	
OK	sdspage_fill_tank_buffer	2 / 2	8	
OK	sdspage_heat_denature_samples	4 / 4	5	
OK	sdspage_prepare_gel_cassette	4 / 4	4	
OK	sdspage_prepare_running_buffer	2 / 2	8	
OK	sdspage_prepare_sample_mix_single_lane	4 / 4	12	
OK	sdspage_recycle_buffer	3 / 3	5	
FAIL	cell_seeding_plate_setup	2 / 4	10	next button not visible after step_validator
FAIL	drug_dilution_setup	1 / 9	9	missing DOM affordance mid-step
FAIL	mtt_plate_reaction	0 / 6	0	next button not visible after step_validator
FAIL	passage_hood_detachment	0 / 9	0	locator.click timeout (visibility)
FAIL	passage_pellet_reseed	1 / 9	2	missing DOM affordance
FAIL	plate_drug_treatment_drug_addition	0 / 9	0	missing DOM affordance
FAIL	sdspage_destain_gel_rock	0 / 3	0	waitForSelector timeout (visible-but-timed wait)

FAIL	sdspage_destain_gel_setup	4 / 6	7	waitForSelector timeout
FAIL	sdspage_image_gel	0 / 3	0	missing DOM affordance
FAIL	sdspage_load_protein_ladder	5 / 6	5	missing DOM affordance
FAIL	sdspage_load_sample_single_lane	2 / 3	5	missing DOM affordance
FAIL	sdspage_run_electrophoresis	0 / 3	0	waitForSelector timeout (visible-but-timed wait)
FAIL	sdspage_stain_gel	2 / 5	4	locator.click timeout
FAIL	trypan_blue_counting	0 / 9	0	missing DOM affordance

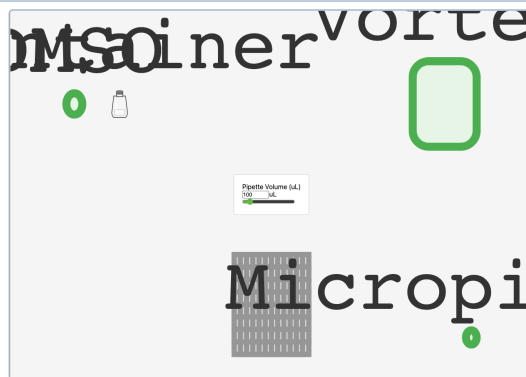
10. Best current examples

From the Round 3 best-5 gallery (subjective ranking after R6 scoreboard). These represent the strongest demo-grade runtime frames available today.

Rank	Scene	Score	Why good
1	staining_bench (after R1 fix)	not scored (post-R1)	Four real volumetric flask SVGs, real Rocking Shaker asset, staining tray clearly labeled
2	sdspace_heat_denature_samples (after R3 label fix)	R6 -4 (initial); R3 cleaner	Labeled Heat Block placeholder, recognizable Microtube rack, real Protein ladder SVG, no overlapping label clusters
3	mtt_solubilization_readout (R3 label CSS variant, after_entry)	R6 -5; click+OSC smoke pass	Only scene with verified click-driven state change: real 96-well plate SVG, DMSO+Waste pair, live Pipette Volume slider
4	mtt_solubilization_readout (after R3 label fix)	R6 -5; click+OSC pass	Same click-through success; hash-marked plate placeholder still legible; Pipette Volume widget is unambiguous focal point
5	electrophoresis_bench (after R1 fix)	not scored (post-R1)	Real Running buffer flask, Protein ladder microtube, Buffer recycle bottle; labeled dashed Power Supply gives semantic structure



Best #5: electrophoresis_bench after R1 fix — labeled dashed Power Supply placeholder gives structure to the rig



Best #3: mtt_solubilization_readout after_entry — only scene with verified click-driven OSC

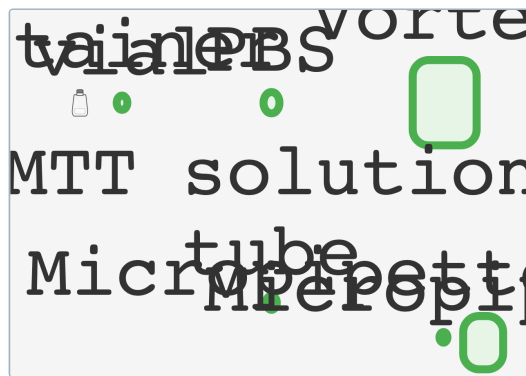
11. Worst current examples

From the Round 3 worst-5 gallery, lowest combined R6 scoreboard rows. These rank the highest-leverage fix targets.

Rank	Scene	Score	Defect callouts	Highest-leverage fix
1	sdspace_attach_lid_and_leads (initial)	-17	5 placeholders, 8 label overlaps, 4 off-page	A1B SVG import (power supply, gel cell, lead-wire) + R7 label-overflow guard
2	mtt_reagent_prep (initial)	-10	4 placeholders, 3 overlaps, 4 off-page	A1B (tube, MTT-solution tube, vortex) + R7 label clamp
3	mtt_solubilization_readout (initial)	-6	2 placeholders incl. hash-mark plate, 2 overlaps, 3 off-page	A1B (real 96-well plate SVG) + R7 DMSO/vortex label width
4	mtt_solubilization_readout (after_entry)	-5	same as #3; slider recovers clarity	R9 propagate click-driven pattern
5	sdspace_heat_denature_samples (initial)	-4	1 placeholder, 2 overlaps, 2 off-page	A1B (real heat-block SVG) + R7 shrink mid-canvas Heat Block label



Worst #1: sdspace_attach_lid_and_leads initial — 5 placeholders, 8 label overlaps, 4 off-page (score -17)



Worst #2: mtt_reagent_prep initial — 4 placeholders, 3 overlaps (score -10)

Cross-cutting: sum placeholder_count=14 across the worst-5; sum (label_overlap + off_page)=33; visible crops sum to zero. Asset import (A1B) is the highest-leverage lever, not crop fixes.

12. What must be scrapped

12.1 `css_native_adapter.ts` (dead spike in the bundle)

`src/scene_runtime/layout/css_native_adapter.ts` (FA Concern 1) is imported by `adapter.ts` but gated by `is_css_native_well_plate_zoom_spike_enabled()` which defaults `false`. The file's own header (line 9) says: "Must be removed or replaced before NEW1 promotion." It has been sitting in `src/` since 2026-05-21 with no plan-of-record to either de-gate or remove. Every production bundle includes it. Remove it; the bundle gets smaller and the ship-risk of an accidental flag flip goes to zero.

12.2 `computeRowSlotSceneLayout` (unwired alternate strategy)

FA Concern 3: `layout_engine.ts` exports `computeRowSlotSceneLayout` (approx lines 700-799). Re-exported by `index.ts` as a named export. Not called in `adapter.ts`, not called in `render/scene.ts`, not called anywhere in the production path. Either wire it, document its purpose, or delete it. As-shipped it just misleads future readers.

12.3 The "experiments will replace the runtime" framing

FF shows experiments-better in 1 of 6 head-to-heads; the experiments precheck verdict shows 0/10 PASS on its own gold templates and 0/110 on its own stress corpus. The CSS-native render path is not a near-promotion path; it is a research artifact that produced useful diagnostics and CSS idioms. Stop scoping work on the assumption that `experiments/css_native_layout/` will be the next render target without a contract amendment (which does not exist and is not approved).

12.4 Showcase concepts and storyboards

FB Feature 7: `showcase/concepts/` `storyboards` (`electrophoresis_setup_walkthrough.html`, `pipette_to_well_storyboard.html`, `selected_well_highlight.html`) demonstrate interaction states that the production dispatch pipeline now handles. `hover_reveal_demo.html` and `selected_well_demo.html` show what CSS-alone visual behavior could be, but production already implements these through protocol dispatch. Obsolete; do not promote.

12.5 Direction A and scratch CSS variants

Per `experiments/css_native_layout/DECISION_MEMO.md`: `bench_a.css`, `bench_b.css`, `bench_d.css`, `bench_diorama.css`, `bench_focusedstage.css`, `bench_gameboard.css` and hood/instrument siblings are gitignored, superseded by Direction B. `spike_css_native_well_plate_zoom.mjs` inlines a JS mirror of the adapter (divergence risk per file header). Remove the spike test once the spike adapter is removed.

13. What should be kept

13.1 Runtime mount path

26/26 protocols mount cleanly with zero `RUNTIME_INITIALIZATION_ERROR` overlays. The mount path through `bundle/entry.ts` → `loader/world.ts` → `render/scene.ts` is the load-bearing achievement of Round 3. Keep.

13.2 Zone-based layout engine

`src/scene_runtime/layout/layout_engine.ts` (`legacyComputeLayout`) is the mined-verbatim production engine, historically active since 2026-05-05, with 8 documented functions. It works; FF shows it beats the CSS-native static templates in 5 of 6 pairings. Keep.

13.3 Dispatch (click and adjust)

`src/scene_runtime/dispatch/click.ts` and `src/scene_runtime/dispatch/adjust.ts` drove 133 successful interactions. The adjust gesture is the continuous set-point primitive (volume sliders, voltage knobs) that the contract requires must not collapse into `click`; it is working. Keep.

13.4 Walker engine

`tests/playwright/walker/engine.mjs` walked 12/26 protocols to completion. Its 14 failures decompose into *walker-instrument* or *scene-wiring* concerns, not runtime mount failures. Improve the walker (test-clock affordance, next-button visibility detection); do not replace it.

13.5 Foundation HTML wiring

The per-protocol HTML shells produced by the build pipeline mount the runtime bundle consistently. No foundation-level wiring bugs surfaced in FG. Keep.

13.6 ns0 fix and chrome stack

The chrome stack (`src/scene_runtime/chrome/*`: prompt, feedback, next button, scene frame, adjust panel) renders without console errors on 24/26 protocols. The two console errors both come from `sdspace_run_electrophoresis` and are non-blocking. Keep the chrome stack; chase the

two errors as a focused fix.

14. What should be promoted from experiments/

Four contract-compatible promotions. None of these require amending PRIMARY_CONTRACT item 3.

14.1 .object-graphic containment wrapper + object-fit: contain

Source: `experiments/css_native_layout/styles/bench.css` lines 155-163 (`.placement { overflow: visible; }`), lines 334-358 (`.object-graphic centering + object-fit: contain`). This is a render-layer CSS rule, not layout engine behavior. PRIMARY_DESIGN.md's no-crop rule currently has *no structural enforcement* in `src/`; the rule text is promoted but the CSS mechanism is not. Promote the wrapper plus the three CSS rules into `src/style.css` and have `src/scene_runtime/render/apply.ts` emit the wrapper. Cost: tiny. Value: every SVG placement gets a safety net against accidental crop regression.

14.2 checkArtworkIntegrity (precheck.mjs)

Source: `experiments/css_native_layout/precheck.mjs` ~lines 1050-1200. Detects `clipped_by_parent` (ancestor overflow != visible) and `aspect_distorted` (rendered aspect ratio deviates from natural by >5%). Port as `tests/playwright/test_artwork_integrity.mjs` running against the canonical 10 templates or against all 26 `dist/*.html` pages. Only automated verification of the no-crop hard rule. The other 11 precheck checks can follow.

14.3 object_footprints.yaml

Source: `experiments/css_native_layout/object_footprints.yaml` (146 entries). Maps object names to footprint size classes. Required to correctly size objects under any future CSS-native render path. Move to `docs/specs/` as authoritative registry or to `generated/` as builder output. Does not change runtime behavior today; unblocks future work.

14.4 The 10 gold stress YAMLS

Source: `experiments/css_native_layout/stress_scenes/gold/gold/*.yaml`. Follow the standard placement/zone vocabulary; register as `tests/content/dev_smoke/ fixtures` (per PRIMARY_SPEC.md "dev_smoke" protocol_type). Exercise adversarial scene compositions no current smoke fixture covers: dual large equipment competing for instrument_station, repeated identical small items, adversarial size mix. No contract amendment needed.

Not promoted (deferred): scene-class CSS dispatch, `templates/dir_b/`, full scorecard infrastructure. These are blocked by PRIMARY_CONTRACT item 3 unless and until the contract is amended (a draft exists at `docs/archive/css_native_layout/new1_primary_contract_item3_amendment_draft.md` but is not approved).

15. Recommended next 1-week plan

Six concrete tracks, sized for parallel execution. Each track is independently completable.

Day 1-2: Promote no-crop CSS mechanism (Track A)

- Add `.object-graphic` wrapper emission to `src/scene_runtime/render/apply.ts`.
- Add three CSS rules to `src/style.css`: `.placement { overflow: visible; }`, `.object-graphic { display: flex; align-items: center; justify-content: center; }`, `.object-graphic img, .object-graphic svg { object-fit: contain; width: auto; height: auto; max-width: 100%; max-height: 100%; }`.
- Re-run `forensic_svg_completeness.mjs`; expect visible-crop count to remain 0 with the structural safety net now in place.

Day 1-2: Remove dead spike (Track B)

- Delete `src/scene_runtime/layout/css_native_adapter.ts`.
- Delete `src/scene_runtime/layout/feature_flags.ts`.
- Remove the gated branch in `adapter.ts`.
- Delete `tests/playwright/spike_css_native_well_plate_zoom.mjs`.
- Either wire or delete `computeRowSlotSceneLayout` (recommendation: delete; if a use case appears later, it is git-retrievable).

Day 2-3: SVG asset gap (Track C, highest leverage on quality)

- Replace generic_outline assets for: `power_supply_off`, `microwave_closed`, `heat_block_closed`, `lightbox_off`, `electrode_module`, `gel_opening_tool`, `kimwipe_pad`, `paper_towel_pad`, `microtube_rack_24`. These are 9 assets driving ~33 of 42 generic_outline hard fails.
- Fix the 6 `well_plate_96` empty_bubble cases: the well plate adapter emits a group but no SVG child; either inline a real plate SVG or have the well plate subpart adapter populate the group.
- Re-run `forensic_svg_completeness.mjs`; target $\geq 95\%$ `real_svg`.

Day 3-4: Promote checkArtworkIntegrity (Track D)

- Port `checkArtworkIntegrity` from `experiments/css_native_layout/precheck.mjs` into `tests/playwright/test_artwork_integrity.mjs`.
- Run against all 26 `dist/*.html` pages.
- Wire into `check_codebase.sh` as a fast gate (skip in `--fast` if needed).

Day 4-5: Walker robustness (Track E)

- Add a test-clock affordance to `src/scene_runtime/render/clock.ts` so visible-but-timed waits (incubation, confluence) can be skipped under walker control. Addresses 5 walker failures.
- Improve next-button visibility detection in `tests/playwright/walker/engine.mjs`. Addresses 2 walker failures.
- Audit "missing DOM affordance" cases (8 protocols); these are scene-wiring gaps where a declared target name has no resolved scene object. Fix per protocol.
- Target: walker pass $\geq 22/26$.

Day 5-6: Stress fixtures and footprint registry (Track F)

- Copy the 10 gold YAMLS from `experiments/css_native_layout/stress_scenes/gold/` into `tests/content/dev_smoke/<name>_check/protocol.yaml`; rename placeholder asset names to existing inventory entries.
- Add a builder pass that registers these as `protocol_type: dev_smoke`.
- Move `object_footprints.yaml` into `docs/specs/OBJECT_FOOTPRINTS.md` (table form) or `generated/object_footprints.yaml`.
- Run mount check on each `dev_smoke` fixture.

Day 6-7: Label policy and console error chase (Track G)

- Apply one of the six worked label policies from `experiments/css_native_layout/showcase/label_policies/` to `src/style.css`. Target: drop runtime label-overlap from 46 to ≤ 10 .
- Investigate the two `sdspace_run_electrophoresis` console errors. Likely a state-machine timing issue with the visible-but-timed wait.

Out of scope for this week

- Any work that requires amending PRIMARY_CONTRACT.md item 3 (scene-class CSS, `templates/dir_b/`, full CSS-native render). If the contract is to be amended, that is its own approval thread.
- Replacing the runtime layout engine. The runtime won 5/6 head-to-heads; there is no evidence that swapping it improves outcomes.
- The full 12-check precheck port. Start with `checkArtworkIntegrity`; promote others as needed.

Closing

The shipping path works at the floor (26/26 mount, 0 crops, 0 placeholders) and fails at quality bars it sets for itself (17.3% SVG hard fails, 46% walker completion, 46 label overlaps). The experiments path failed earlier and harder (0/10 PASS, 0/110 stress) but produced four genuinely useful artifacts that should be promoted. The salvage suspicion is closed: `src/scene_runtime/layout/` was mined from production code, not from broken archives. The dead spike and the unwired alternate strategy should be removed; the runtime should be improved, not replaced.

